Computational Complexity

1. We learnt about automaton

   - DFA
     - Regular language
     - Regular Expressions
   - PDA
     - CFLs
     - CFG
   - Turing Machines

- Decidable problem
- Undecidable problems.

We will focus on

Decidable problems

Even when a problem is decidable, if the problem requires an inordinate amount of time and memory, it may not be solvable in practice.

computational complexity is an investigation of time, memory or other resources required for solving computational problems.

Definition 7.1

Let M be a deterministic Turing Machine that halts on all inputs. Thus, the running time or time complexity of M is the function $f: \mathbb{N} \longrightarrow \mathbb{N}$, where $f(n)$ is the maximum number of steps that M uses on any input of length n.

If $f(n)$ is the running time of M, we say that M runs in time $f(n)$ and M is an $f(n)$ time Turing Machine.

$$f(n) = n + 5n^2 + 3$$

$$f_1(n) = 2^n + 100n + 20$$

Definition

Let f and g be functions, $f, g : \mathbb{N} \longrightarrow \mathbb{R}^+$, we say that $f(n) = O(g(n))$

$$\exists c > 0, n_0 \geq 0 : \forall n \geq n_0 : f(n) \leq c \cdot g(n)$$

when $f(n) = O(g(n))$, we say that $g(n)$ is an upper bound for $f(n)$

or

$g(n)$ is an asymptotic upper bound on $f(n)$

or

f is less than or equal to g if we disregard the differences up to constant factor

$$f(n) = 2n^2 + n + 500$$

$$g(n) = n^2$$

$$f(n) = O(n^2)$$

$$f(n) = 5n^3 + 2n^2 + 11n + 5$$

$$g(n) = n^3$$

$$f(n) = O(n^3)$$

$$c = 23 \quad , \quad n \geq 1$$

$$23n^3 \geq 5n^3 + 2n^2 + 11n + 5$$

Little-o-notation

Let f and g be functions, $f, g: \mathbb{N} \longrightarrow \mathbb{R}^+$, say $f(n) = o(g(n))$ if

$$\lim_{n \to \infty} \frac{f(n)}{g(n)} = 0$$

In other words, $f(n) = o(g(n))$ means for any constant $c > 0$ a number n_0 exists s.t

$f(n) < g(n) \quad \forall n \geq n_0$

$$f(n) = n^3 + 2n^2 + 5$$

$$h(n) = n^3$$

$$f(n) = o(n^3) \quad \times$$

$$\lim_{n\to\infty} \frac{n^3+2n^2+5}{n^3} = \underbrace{\lim_{n\to\infty} 1}_{\to 1} + \underbrace{\lim_{n\to\infty} \frac{2}{n}}_{\to 0} + \underbrace{\lim_{n\to\infty} \frac{5}{n^3}}_{\to 0}$$

$$\times$$

$$f(n) = o(n^4)$$

$$\lim_{n\to\infty} \frac{n^3+2n^2+5}{n^4} = 0$$

we can classify languages based on their time requirements.

Definition

$$\text{TIME}(t(n)) = \{ L \mid L \text{ is a language decided by an } O(t(n)) \text{ time TM} \}$$

Class of P

$$P = \bigcup_{k} TIME(n^k)$$

The class of languages P contains all decision problems that can be solved in polynomial time on a deterministic single-tape turing machine.

All languages that can be decided in polynomial time is contained in this class.

Note that problems that can be solved in $O(n^{100})$ are in P, even though in practice, this is a practically difficult problem to solve.

$$PATH = \{\langle G,s,t\rangle \mid G \text{ is a directed graph with nodes } s \text{ and } t \text{ and there is a path from } s \text{ to } t\}$$

$PATH \in P$